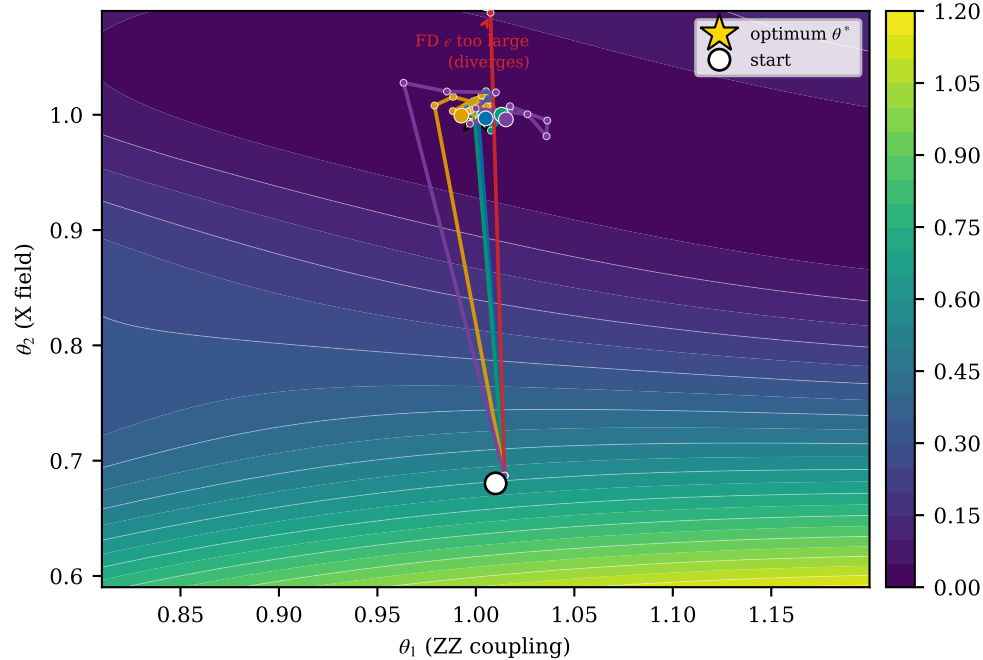


F-loop —  $\varepsilon$ -free PSR/NSR reliably recover an interior  $\theta^*$  on a sloppy IQS valley; finite differences need an  $\varepsilon$  they cannot tune without the answer: too-large  $\varepsilon$  FAILS (truncation  $\frac{\varepsilon^2}{6}|C'''|/\mu_{\text{soft}}$ ), too-small  $\varepsilon$  is UNRELIABLE ( $\delta/\varepsilon$  noise, wide IQR), and even the best  $\varepsilon$  only reaches an uncertifiable  $\delta$ -floor  
 (2q TFIM,  $P=2$ ; compiled to machine-native segments, emulated under T4;  $T/T_2^* = 0.15$ ; 20 seeds, median  $\pm$  IQR;  $w=0.250$ ,  $\mu_{\text{soft}}=0.54$ )

(a) sloppy IQS valley:  $\varepsilon$ -free PSR/NSR reach  $\theta^*$ ; FD too-large fails, small- $\varepsilon$  is unreliable



(b) convergence: PSR/NSR reach  $\theta^*$  reliably; every FD  $\varepsilon$  either fails, floors, or is unreliable

